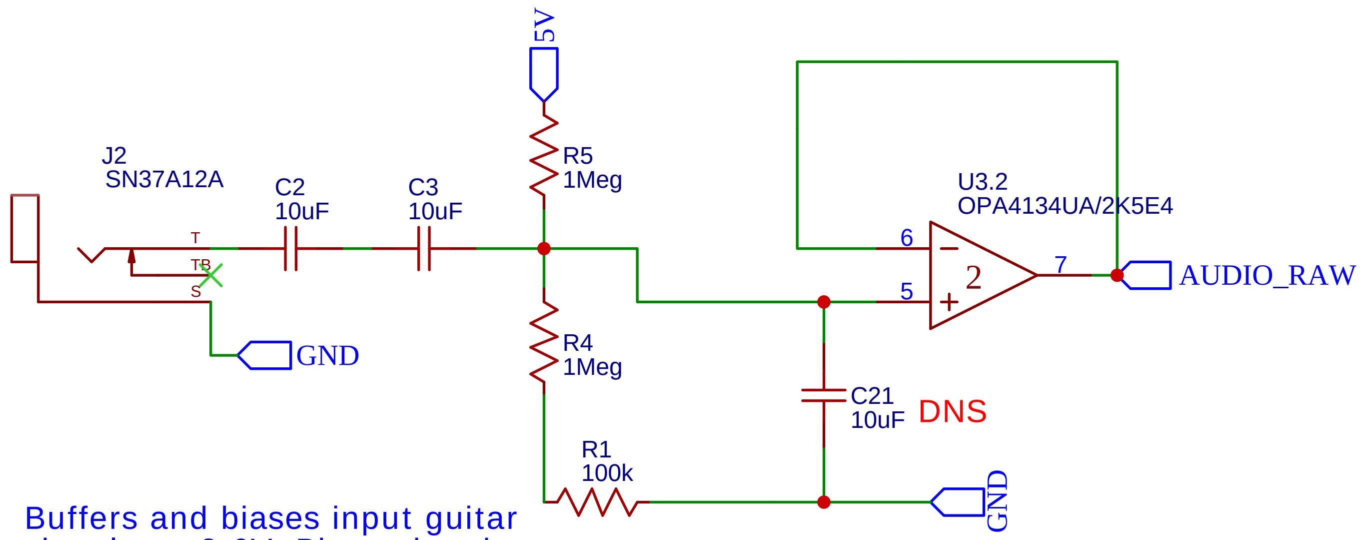
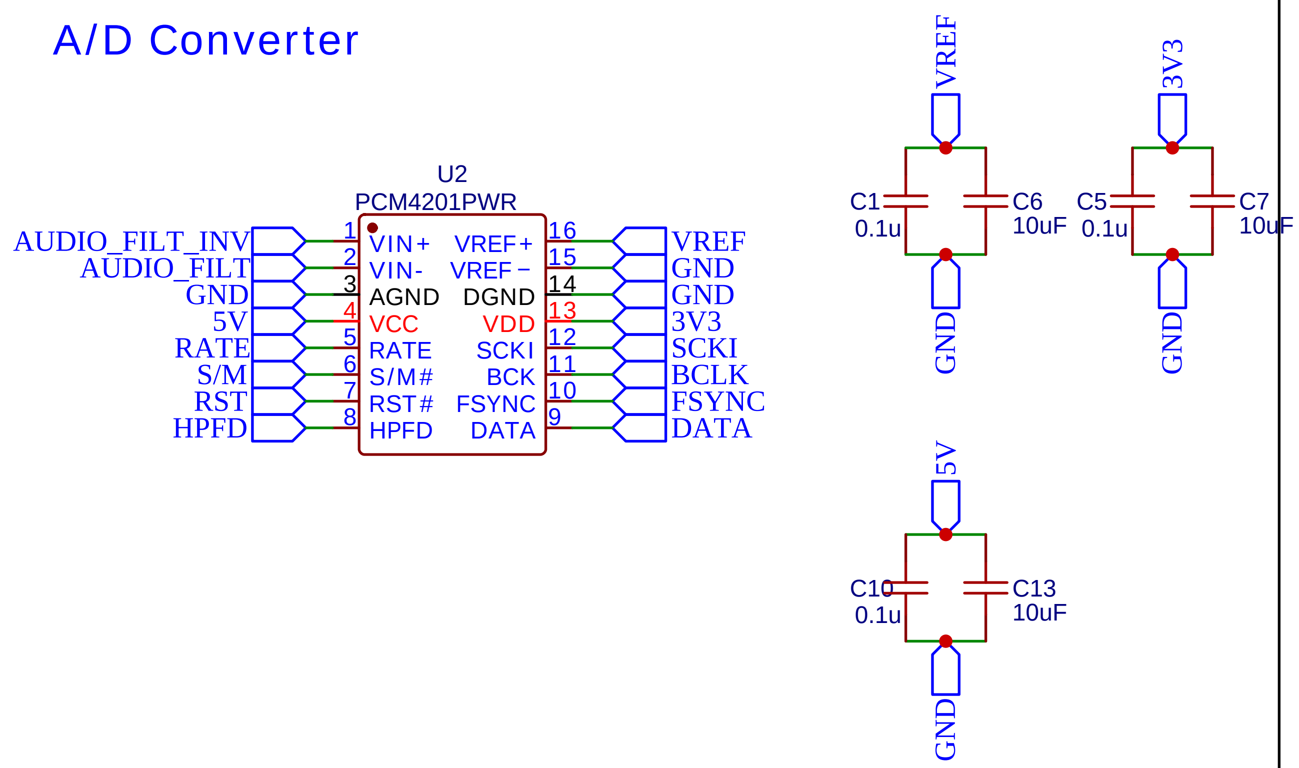
Audio Jack (1/4)

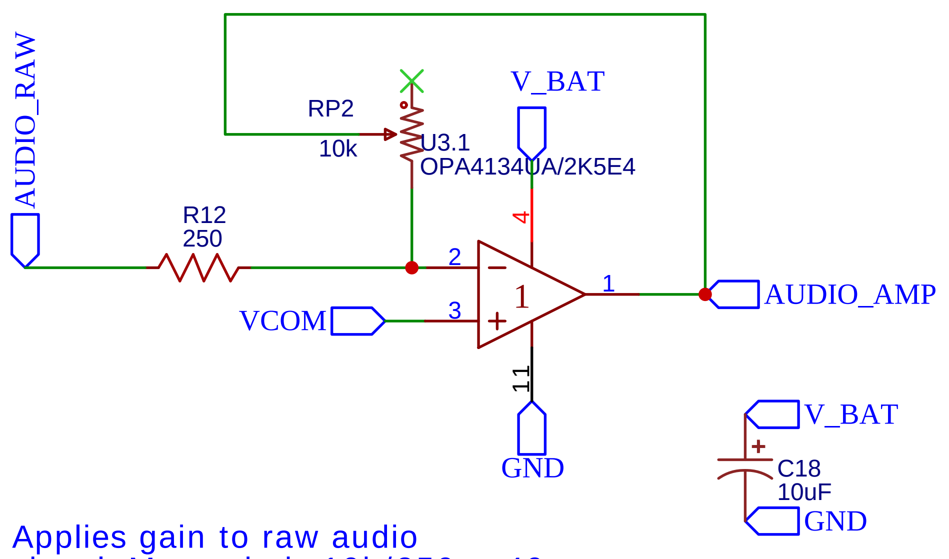


Buffers and biases input guitar signal to ~2.6V. Bias point chosen due to common mode considerations. Cap values chosen to provide good DC attenuation and little attenuation of 40hz+ frequencies.

A/D Converter

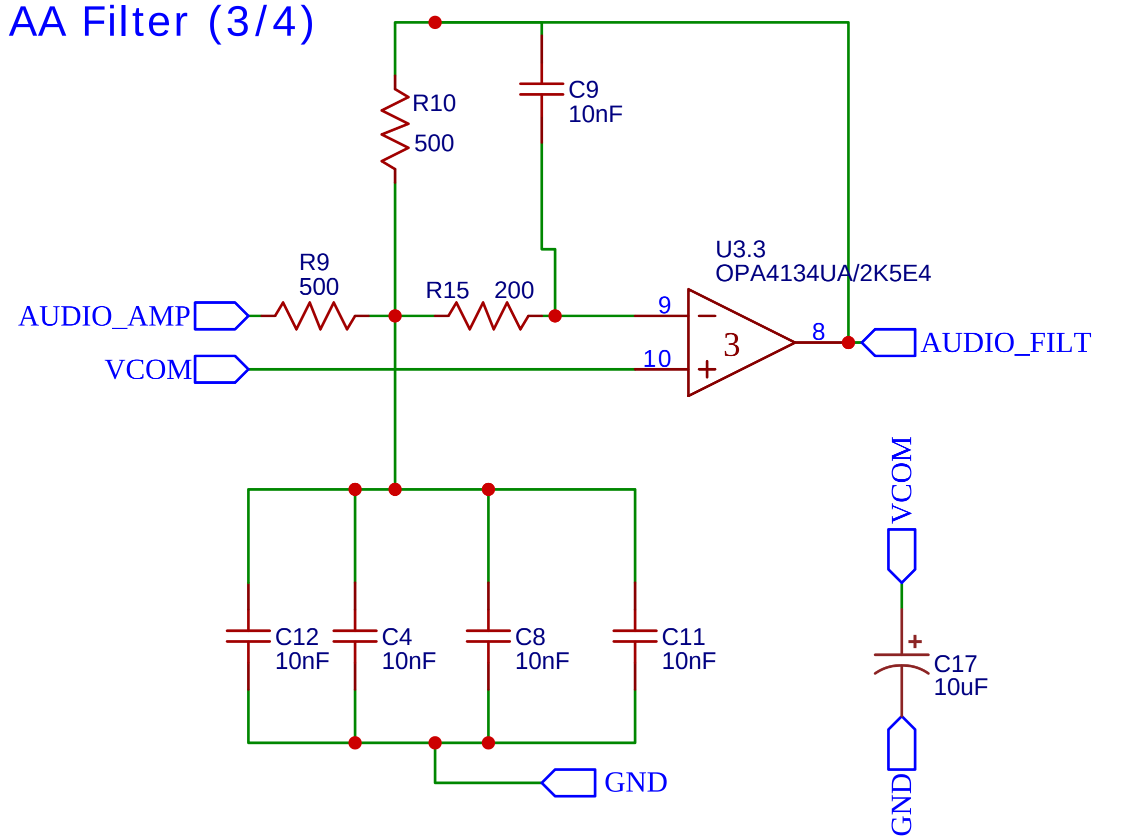


Gain Stage (2/4)

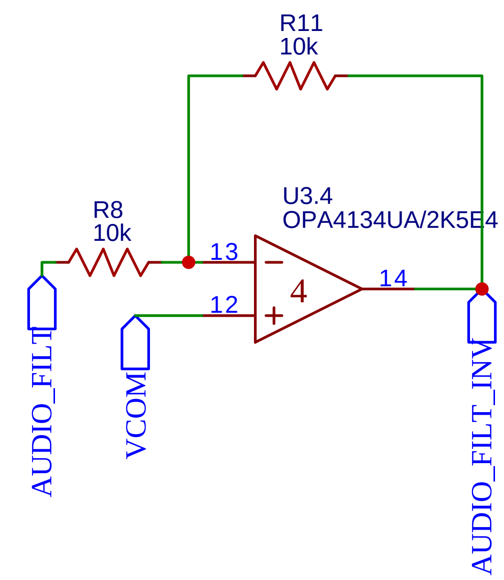


Applies gain to raw audio signal. Max gain is 10k/250 = 40. Potentiometer can be adjusted to selected gain.

AA Filter (3/4)

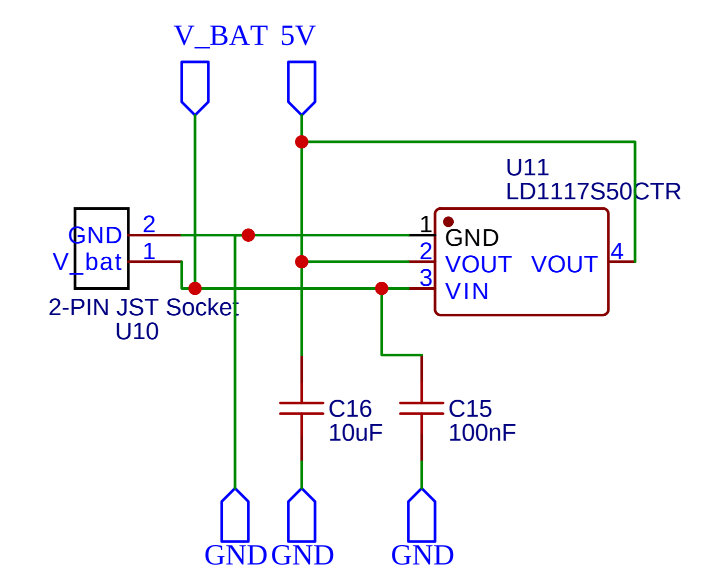


Inverting Stage (4/4)

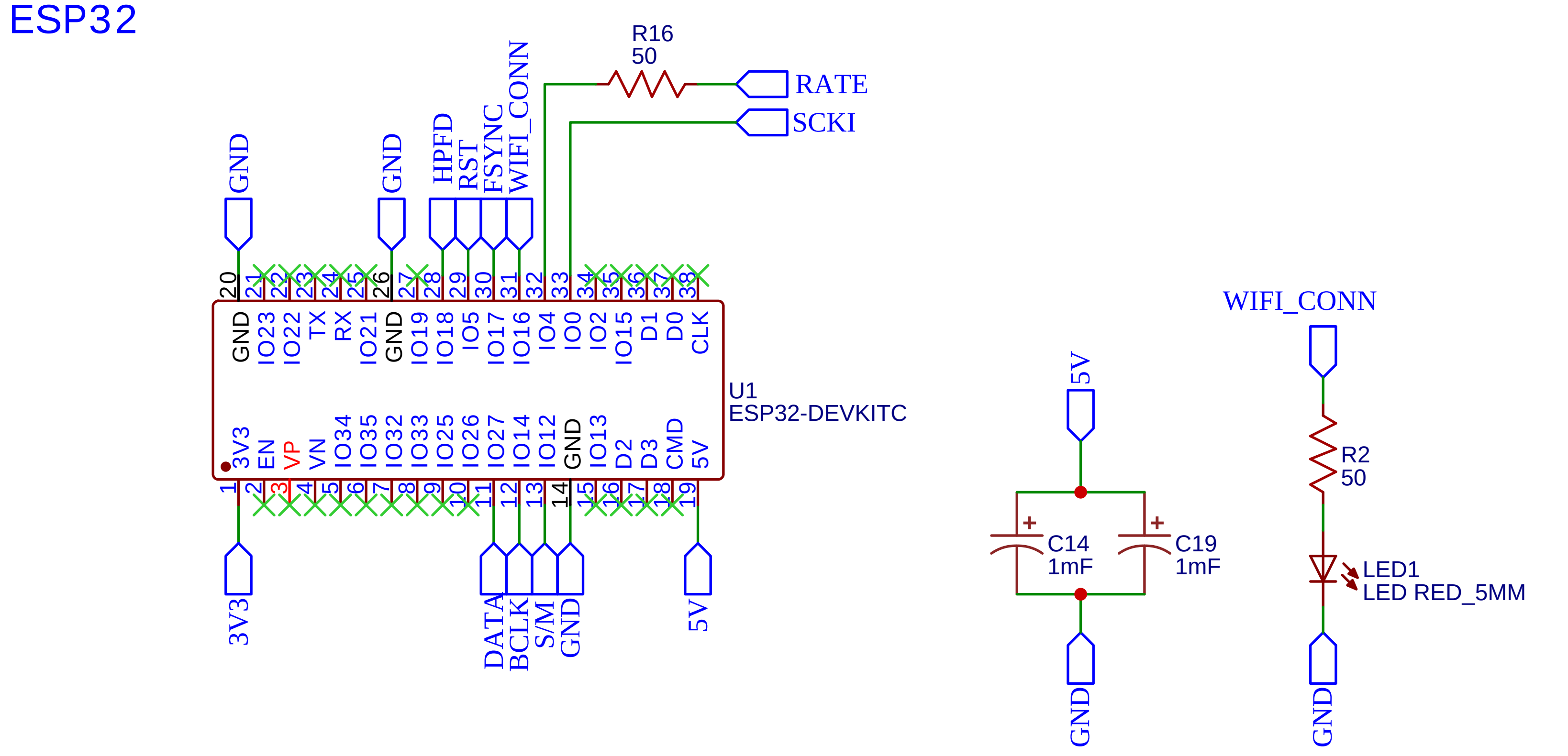


TITLE: Sheet_1	REV: 1.0
Company: Your Company	Sheet: 1/1
Date: 2025-05-15 Drawn By: rahulahoop	

Power Supply



ESP32



Vf = 2.1-2.6V
Ima=14-24mA

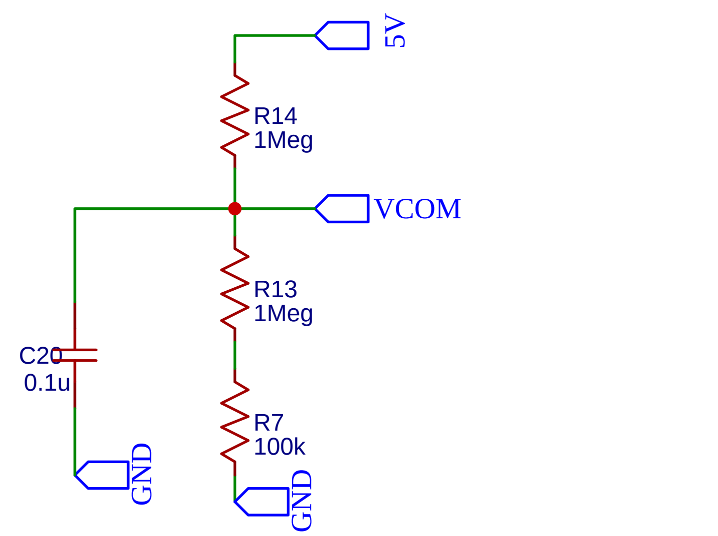
The following pins have been verified to work as I2S TX:
GPIO 12
GPIO 14
GPIO 27

GPIO 17
GPIO 5
GPIO 21

Pins 4, 5, 18, 12
verified as output GPIOs.

GPIO 32 is part of
ADC Unit 1, which works
during Wifi communication

VCOM



TITLE: Sheet_2	REV: 1.0
Company: Your Company	Sheet: 1/1
Date: 2025-06-21 Drawn By: rahulahoop	